

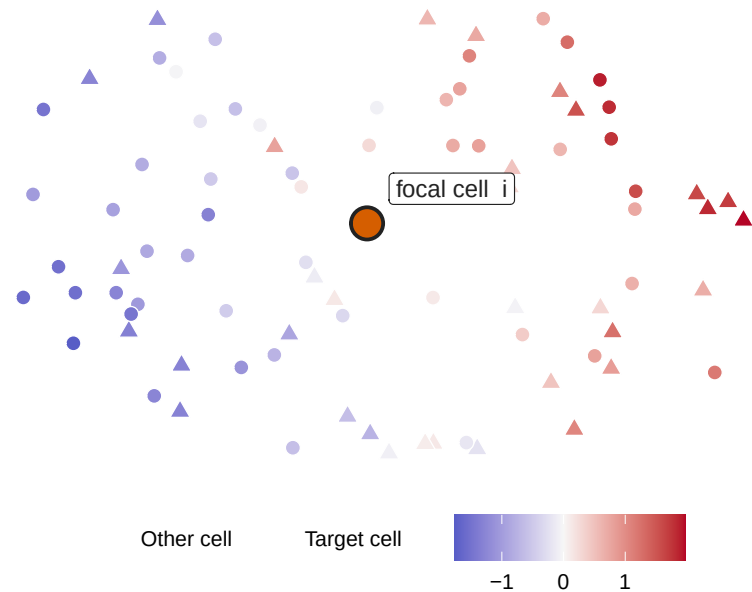
Construction of the multiscale cellular-context representation

Single-cell embeddings are averaged at two spatial scales, concatenated, and regularized across nearby target cells

A

Cell-level embedding

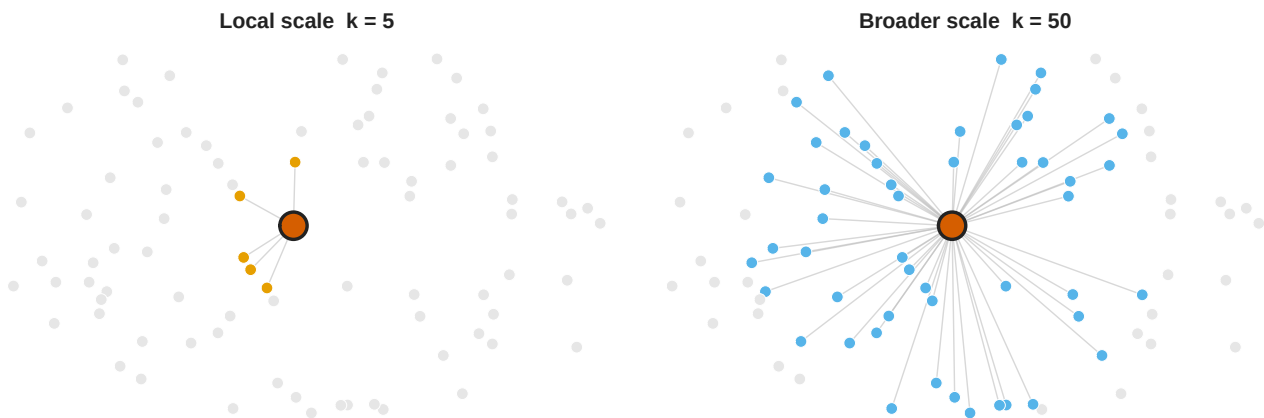
Each cell carries a vector of PCA scores



B

Multiscale spatial averaging

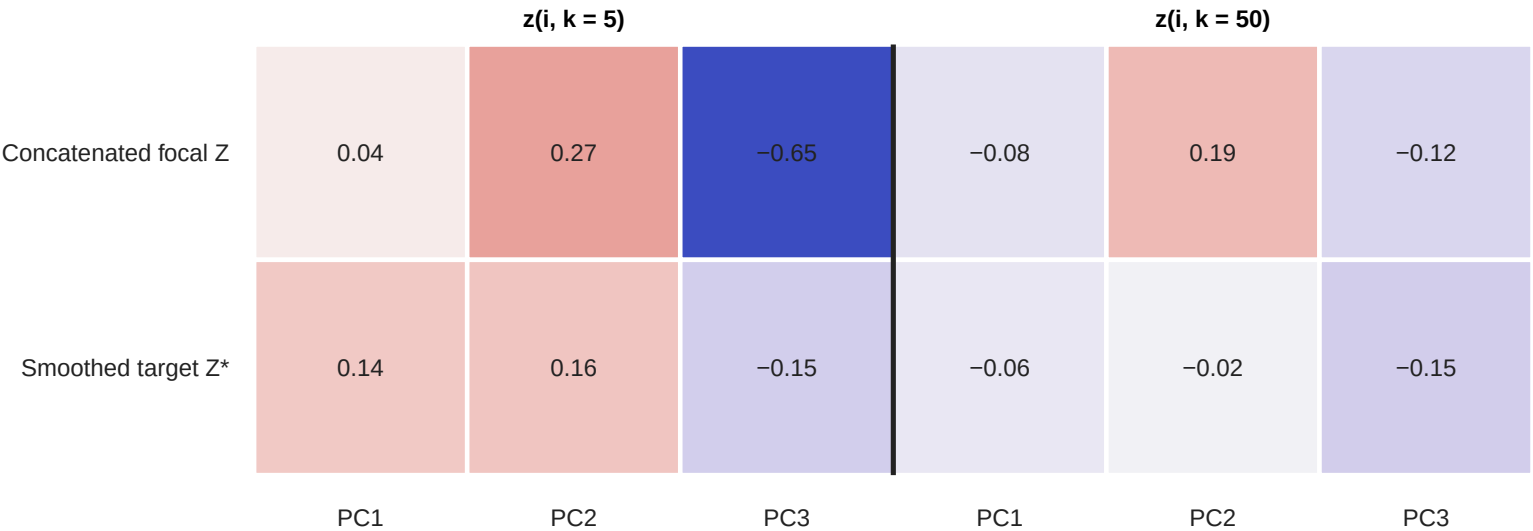
PCA scores are averaged separately at each scale



C

Concatenation and smoothing

The two scales remain separate blocks in Z



D

Target-cell smoothing

Multiscale Z vectors are averaged over nearby targets

